

Posets

Hiten Tandon

1 Notation

$\mathbb{N}$ = Natural Numbers

$\mathbb{P}$ = Positive Integers

$\mathbb{Z}$ = Integers

$\mathbb{R}$ = Real Numbers

$\mathbb{C}$ = Complex Numbers

$\mathbf{R}$ = Commutative Unital RING

$\mathcal{P}$ = Set of all Prime numbers

$[a, b] = \{x \in \mathbb{Z} \mid a \leq x \leq b\}$

$[a] = [1, a]$

$\xrightarrow{\sim}$ = Used to represent a bijective function (one-one mapping)

$\xrightarrow[\phi]{\sim}$ = Used to show the bijective mapping under the function ϕ

$\hat{0}$ = Smallest element of the Poset

$\hat{1}$ = Greatest element of the Poset

$\leq_X$ = less than or equal, in the poset X .

$n(\dots)$ = cardinality of the expression in the parenths.

2 Partially Ordered Sets / Posets

A partially ordered set also known as a Poset is a pair of a set P and a relation $\leq: P^2$ s.t it satisfies:

1. $\forall x \in P. x \leq x$ (Reflexivity)
2. $\forall x, y, z \in P, x \leq y, y \leq z. x \leq z$. (Transitivity)
3. $\forall x, y \in P, x \leq y, y \leq x. x = y$. (Anti-symmetry)

Common examples of Posets include

1. all subsets of $(\mathbb{C}, \leq)$
2. $(\mathbb{P}, |)$
3. $\forall a, b \in \mathbb{Z}. [a, b]$
4. $\forall X : \text{Set}. (2^X, \subseteq)$
5. $(\mathbb{R}^2, \{(x, y), (x', y') \in \mathbb{R}^2 \mid x \leq x', y \leq y'\})$

Chains and Anti-chains

A Poset $(P, \leq)$ is called a chain if $\forall x, y \in P, x \leq y$ or $y \leq x$. Common examples of this would include $\mathbb{N}, \mathbb{P}, \mathbb{R}$ and their subsets.

A Poset $(P, \leq)$ is called an anti-chain if $\forall x, y \in P, x \leq y \iff y \leq x$. A common example of an antichain would be the set of all human languages, or programming languages.

2.1 Hasse Diagrams

A common way to graphically represent a Poset is by using Hasse diagrams. A Hasse diagram is a graphical representation of a Poset where the following conditions are met:

1. Vertices of the graph represent an element of the poset P
2. There is an edge from x to y if x covers y .

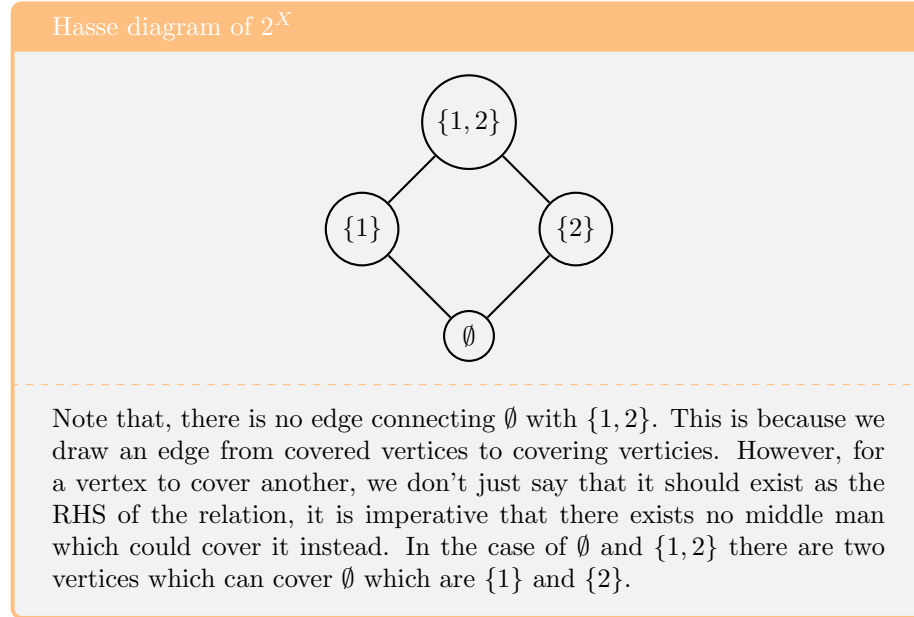
Cover here means the following: If $\forall x, y \in P. \nexists z \in P, z \leq x, y \leq z, x \neq z, y \neq z$. then x is said to cover y .

Generally, Hasse diagrams are drawn without arrows, by placing the covering element physically above the covered element.

For example: Consider the Poset $(2^{[2]}, \subseteq)$ The containment relation can be listed as:

X	$Y \in 2^{[2]} \text{ s.t. } X \subseteq Y$
$\emptyset$	$\emptyset, \{1\}, \{2\}, \{1, 2\}$
$\{1\}$	$\{1\}, \{1, 2\}$
$\{2\}$	$\{2\}, \{1, 2\}$
$\{1, 2\}$	$\{1, 2\}$

This would be drawn in a Hasse diagram as follows:

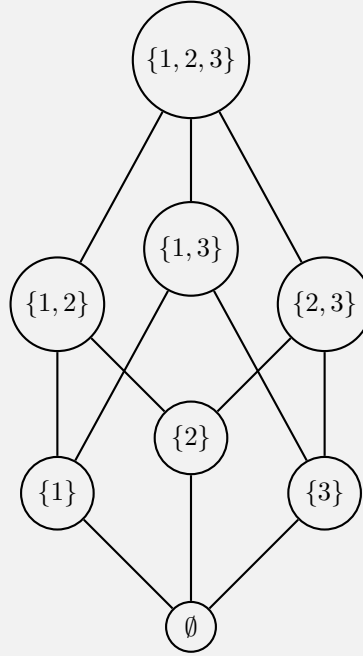


Similarly now let's look at Hasse diagram for $([3], \subseteq)$

We get the table as:

X	$Y \in 2^{[3]} \text{ s.t. } X \subseteq Y$
$\emptyset$	$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$
$\{1\}$	$\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}$
$\{2\}$	$\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}$
$\{3\}$	$\{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$
$\{1, 2\}$	$\{1, 2\}, \{1, 2, 3\}$
$\{1, 3\}$	$\{1, 3\}, \{1, 2, 3\}$
$\{2, 3\}$	$\{2, 3\}, \{1, 2, 3\}$
$\{1, 2, 3\}$	$\{1, 2, 3\}$

Hasse diagram of 2^X



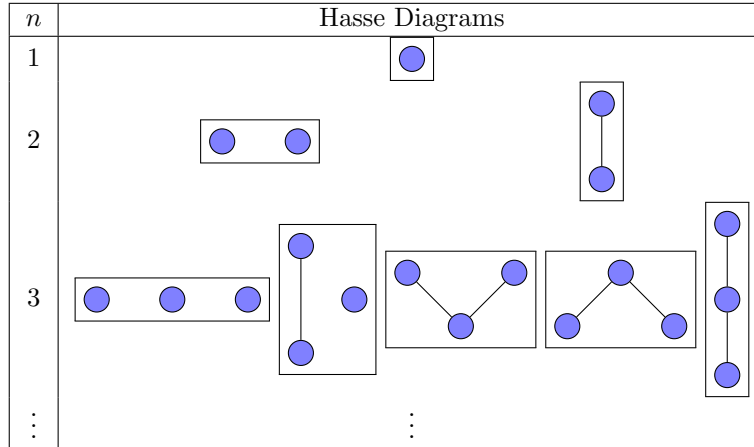
2.2 Isomorphisms of Poset

An isomorphism is represented using the symbol $\cong$ and two posets P and Q are considered isomorphisms if there exists an order transferring bijection from P to Q .

Basically, $(P, \leq) \cong (Q, \leq) \iff \exists \phi: P \xrightarrow{\sim} Q. \forall x, y \in P, x \leq y \implies \phi(x) \leq \phi(y)$

A common example of isomorphic posets includes $([n], \leq) \cong ([a, a+n-1], \leq)$ the bijection is $x \xrightarrow[\phi]{\sim} a+x-1$

Note that, for all pairs of isomorphic posets, we can trivially say that their Hasse diagrams will also be isomorphic. Using this, we can create a beautiful representation showing all the possible isomorphism classes of partially ordered sets, depending on the number of elements that they have.



2.3 Maximal, Minimal, Greatest and Least elements

For a Poset $(P, \leq)$ we define the following:

$x \in P$ is maximal $\iff \forall y \in P, x \leq y. y \leq x$.

$x \in P$ is minimal $\iff \forall y \in P, y \leq x. x \leq y$.

$x \in P$ is greatest $\iff \forall y \in P. y \leq x$.

$x \in P$ is least $\iff \forall y \in P. x \leq y$.

Trivially, we can say that, $\forall (P, \leq), P \neq \emptyset, n(P) < \infty. \exists \text{maximal}(P), \text{minimal}(P)$. Similarly, we can say that, there maybe at most a single greatest and least elements of a poset. Not necessary for there to be one, but if there is, then there can only be one.

2.4 Induced Subposets

Let $(P, \leq)$ and $(Q, \leq)$ be Posets s.t.

$Q \subset P$ and $\forall q_1, q_2 \in Q. q_1 \leq_Q q_2 \iff q_1 \leq_P q_2$ then Q is called an induced subposet of P .

Induced subposets are really useful for studying various isomorphism classes and their relations (how one could go to another)

For example, consider the poset $(\mathbb{P}, |)$ which is poset of Positive integers, ordered by divisibility. Subposets of $(\mathbb{P}, |)$ of the form

$p, q \in \mathbb{P}, \gcd(p, q) = 1, p \neq q. (\{p^a \cdot q^b : a, b \in \mathbb{N}\}, |)$

are isomorphic to the poset $(\mathbb{N}^2, \leq)$ where $(a, b) \leq_{\mathbb{N}^2} (c, d) \iff a \leq_{\mathbb{N}} c, b \leq_{\mathbb{N}} d$.

More generally, if we fix n distinct, pairwise coprime numbers: $(c_i)_{i=1}^n$.

Then, let

$(C_n, |)$ be $\left\{ \prod c_i^{k_i} : \forall i. k_i \in \mathbb{N} \right\}$ and,

$(\mathbb{N}^n, \leq)$ be a poset where $\leq$ is defined by $(a_i)_{i=1}^n \leq (b_i)_{i=1}^n \iff \forall i \in [n]. a_i \leq b_i$.

Then $(C_n, |) \cong (\mathbb{N}^n, \leq)$

2.5 Incidence Algebra

Let $(P, \leq)$ be a Poset such that $\forall x, z \in P, n(y \in P : x \leq y \leq z) < \infty$

Then we define

$A(P) := \{f : P \times P \rightarrow R \mid f(x, y) \neq 0 \implies x \leq y\}$ and,

$*$: $A(P) \times A(P) \rightarrow A(P)$ as,

$$(f * g)(x, y) := \sum_{x \leq z \leq y} f(x, z) \cdot g(z, y)$$

Theorem 2.1. $*$ is associative

Proof. To prove that $\forall (P, \leq). \forall f, g, h \in A(P). (f * g) * h = f * (g * h)$.

Let's start by expanding the LHS and RHS.

LHS:

$$((f * g) * h)(x, z) = \sum_{x \leq y \leq z} (f * g)(x, y) \cdot h(y, z) = \sum_{x \leq y \leq z} \sum_{x \leq w \leq y} f(x, w) \cdot g(w, y) \cdot h(y, z)$$

RHS:

$$(f * (g * h))(x, z) = \sum_{x \leq y \leq z} f(x, y) \cdot (g * h)(y, z) = \sum_{x \leq y \leq z} \sum_{y \leq w \leq z} f(x, y) \cdot g(y, w) \cdot h(w, z)$$

Notice that

1. On the LHS, we're traversing all the quadruplets (x, y, w, z) such that $x \leq w \leq y \leq z$
2. On the RHS, we're traversing all the quadruplets (x, w, y, z) such that $x \leq y \leq w \leq z$

Which simply means that, if we swap w and y on RHS it will be equivalent to the LHS.

□